

Benchmarking **Parameter-Space** vs. Activation-Space Decompositions for Mechanistic Interpretability

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MOTIVATION

Decomposing Networks into Mechanisms

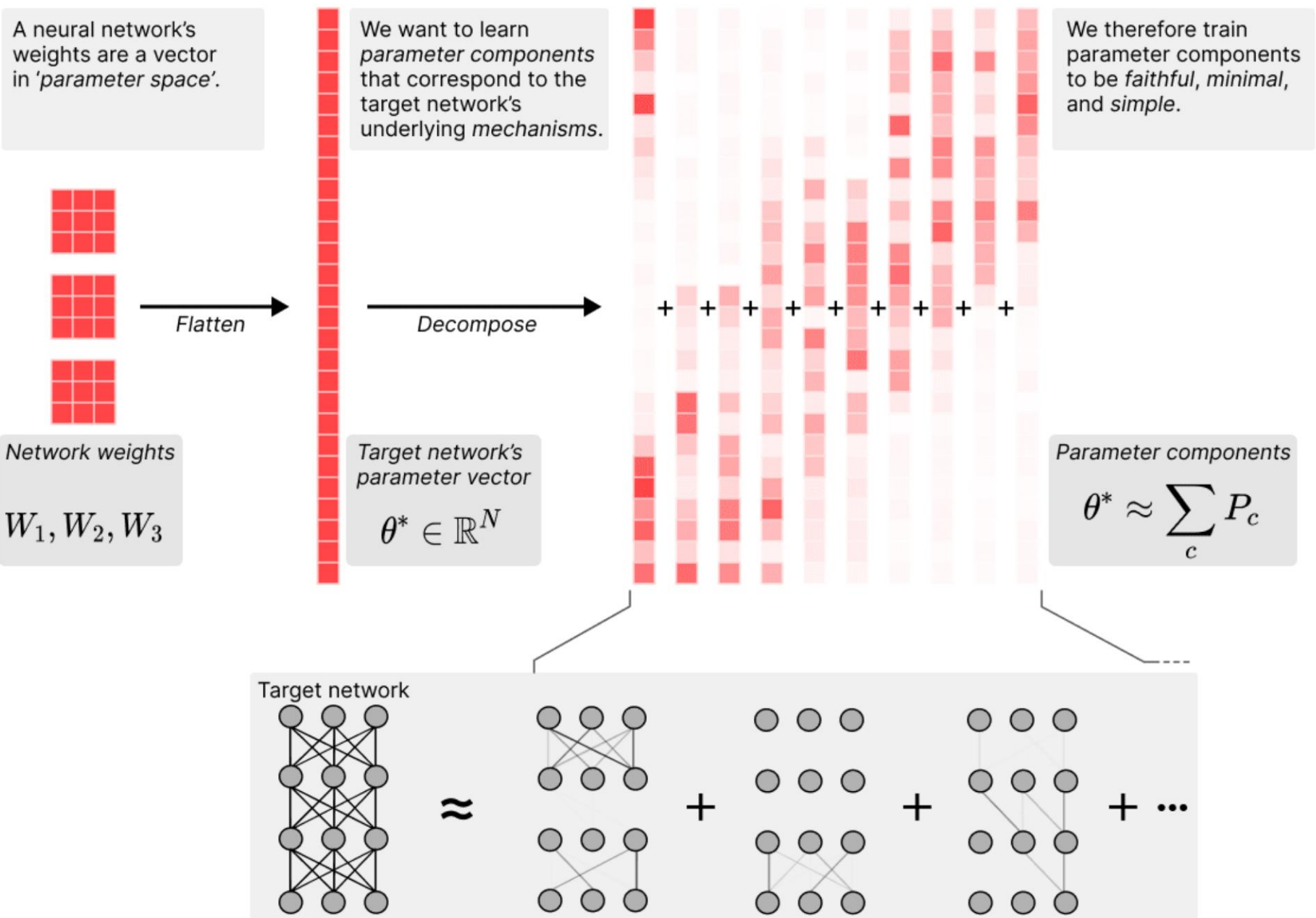
Language models learn algorithms we can't design directly. Mechanistic interpretability aims to reverse-engineer these networks, but naïve units — neurons, attention heads, layers — don't map to individual, interpretable computations.

Activation-Space SAEs, Transcoders, CLTs, MOLTs	Parameter-Space SPD (this work)
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Activation methods approximate the network using functions of a **different functional form**, making it hard to relate their accounts to the actual parameters doing the computation. They also suffer from **feature splitting**, **absorption**, and unreliable **mechanistic faithfulness**. Parameter decomposition methods instead decompose the network's own parameters into components that sum to the original weights.

METHOD

Stochastic Parameter Decomposition (SPD)



SPD decomposes weight matrices into rank-one subcomponents trained to be faithful, minimal, mechanistically faithful, and simple.

Each weight matrix is decomposed: $W \approx \Sigma_c u_c v_c^T$. A learned **causal importance (CI)** function gates which subcomponents are active per input. Key training ingredients:

- Stochastic + adversarial ablations:** Test whether components can be partially ablated in any combination, ensuring mechanistic faithfulness.
- Frequency minimality:** Encourages each subcomponent to activate on as few datapoints as possible, preventing polysemanticity.

Parameter-faithful Components sum to original weights	Minimal Few components active per input
Mech. faithful	Simple

SETUP

Experimental Design

Target model: 4-layer, 67M param decoder-only transformer ($d_{\text{model}}=768$, 6 heads) trained on The Pile. SPD decomposes **24 weight matrices** (6 per layer) into **39,936 total subcomponents**.

Baselines: Transcoders (cascading, parallel, independent), Cross-Layer Transcoders (CLTs), neurons. All activation methods use 32k dictionary size. We evaluate under three protocols: when replacing all MLPs, each encoder either sees the residual stream modified by earlier reconstructions (*cascading*) or the original clean stream (*parallel*); we also test replacing a *single MLP* at a time.

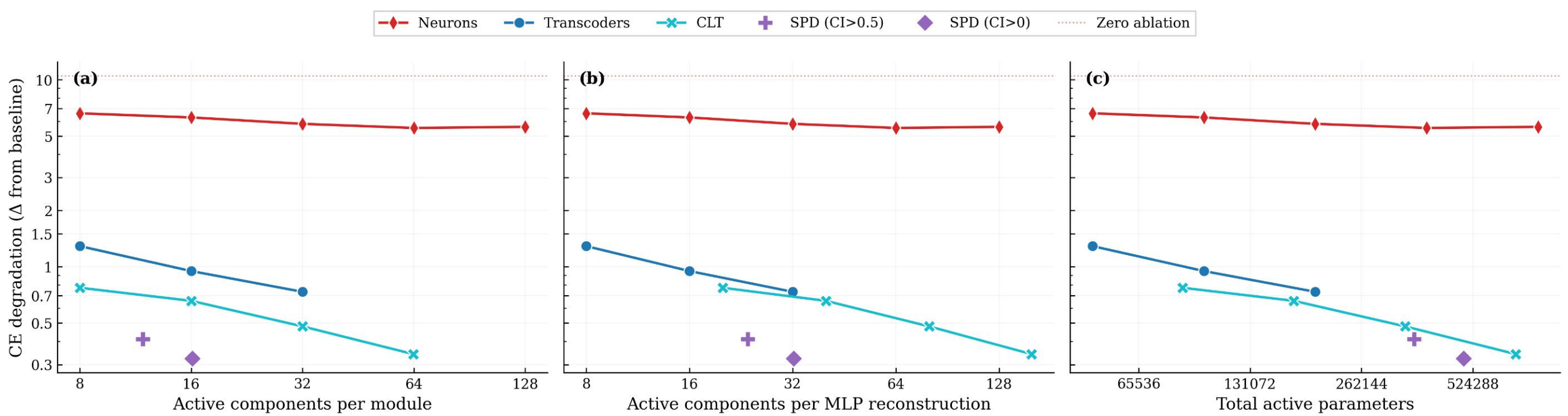
Comparing sparsity requires care: a CLT feature writes to all layers simultaneously; a transcoder feature affects one layer; SPD subcomponents are scoped to individual weight matrices. We report under **three LO normalizations** to ensure conclusions aren't artifacts of counting.

REFERENCES

[1] Braun et al. (2025). Interpretability in parameter space. arXiv:2501.14926 [2] Bushnaq et al. (2025). Stochastic Parameter Decomposition. arXiv:2506.20790 [3] Dunefsky et al. (2024). Transcoders find interpretable LLM feature circuits. arXiv:2406.11944 [4] Lindsey et al. (2024). Sparse crosscoders. Anthropic Research. [5] Leask, Bussmann et al. (2025). SAEs do not find canonical units. arXiv:2502.04878 [6] Sharkey et al. (2025). Open problems in mechanistic interpretability. arXiv:2501.16496

RESULT 1

SPD Achieves a Better Sparsity–Accuracy Tradeoff



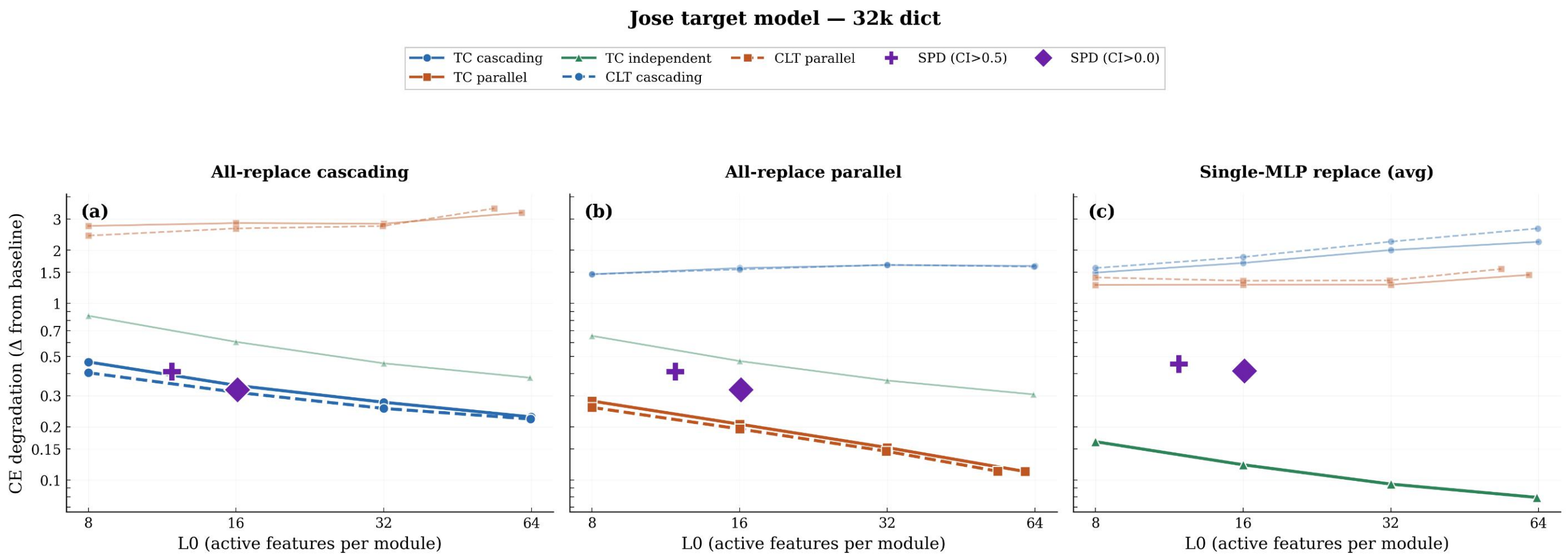
(a) Active components per module. (b) Per MLP reconstruction. (c) Total active params. SPD (purple) Pareto-dominates under all three sparsity measures. Two SPD points correspond to different CI thresholds: $CI>0.5$ (stricter, fewer active components) and $CI>0$ (lenient). Dashed line = zero ablation. Lower is better.

We first compare SPD against transcoders and CLTs trained with the standard **MSE reconstruction loss** on intermediate activations. SPD achieves lower CE degradation at comparable sparsity, and the ordering is **consistent across all three normalizations**. Even with the lenient $CI>0$ threshold, SPD remains well below the activation-based methods.

This advantage may stem from SPD's end-to-end training on the model's output distribution, vs. the local layer-wise MSE objective. To control for this, we train all activation methods e2e with **KL divergence** on output logits, matching SPD's objective ↓

RESULT 2

E2E Activation Methods Overfit to Their Training Mode



(a) **Cascading:** replace all 4 MLPs; each encoder sees the residual stream after earlier reconstructions (errors accumulate). (b) **Parallel:** replace all 4 MLPs; each encoder sees the original clean stream (errors don't propagate). (c) **Single-MLP:** replace one MLP at a time, averaged over layers. Highlighted lines = training mode matches eval mode; faded = mismatched. SPD (purple) is stable across all three protocols.

We define $\delta = \text{CE}_{\text{patched}} - \text{CE}_{\text{baseline}}$ (CE degradation from replacing MLPs with reconstructions; lower is better). A faithful decomposition should perform equally well regardless of evaluation protocol — if an encoder has truly learned its MLP's function, its output should be correct whether or not earlier layers' MLPs have also been replaced. We test **matched-mode** (trained and evaluated in the same protocol) against **cross-mode** (trained in one, evaluated in another).

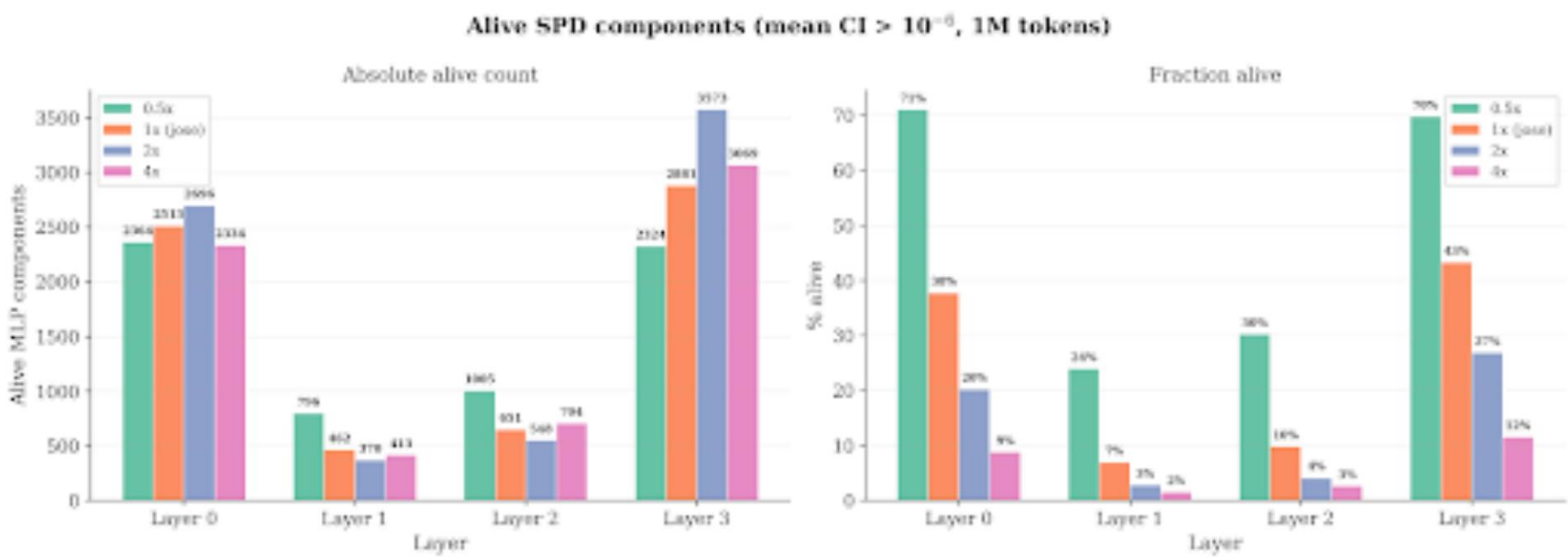
Activation methods show **severe mode-mismatch brittleness**: parallel-trained transcoders reach $\delta \approx 0.14$ in matched mode but degrade to $\delta \approx 2.9-3.5$ under cascading evaluation — comparable to zero ablation (dashed line). The pattern is symmetric: cascading-trained models fail under parallel evaluation. This indicates e2e-trained encoders learn **compensatory strategies** that jointly account for the MLP's computation *and* systematic reconstruction errors from earlier layers.

Independently-trained transcoders are the most robust activation method in single-MLP replacement ($\delta \approx 0.08-0.19$), but still degrade when all four are inserted simultaneously ($\delta \approx 0.38-0.85$).

Key insight: SPD achieves $\delta \approx 0.32-0.40$ regardless of protocol. An SPD reconstruction is not a different function approximating the MLP — it is a **subset of the MLP's own computations**. Matched-mode methods achieve lower raw δ , but this is the expected cost of a more faithful decomposition.

RESULT 3

Feature Splitting — Component Stability

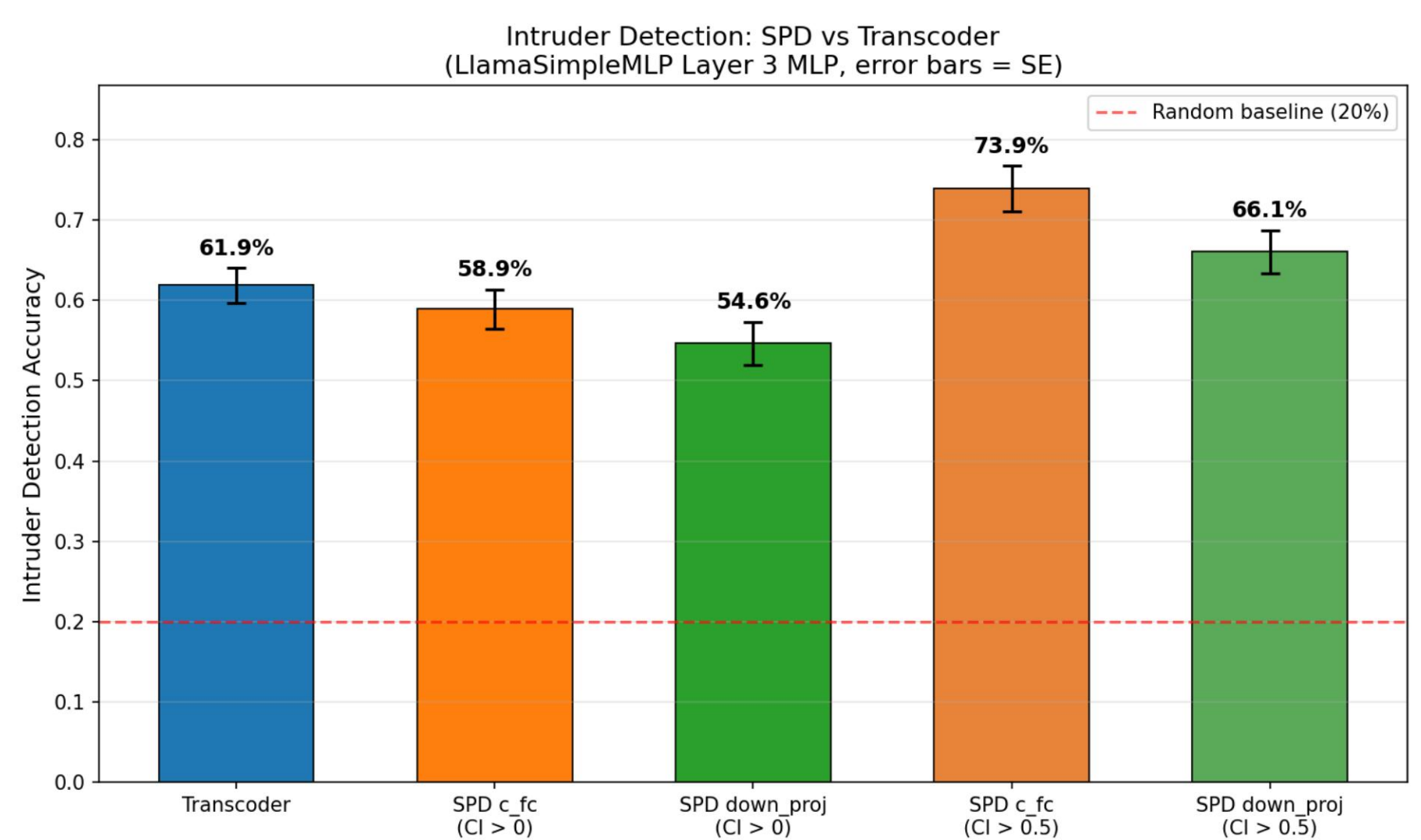


Alive SPD components (mean $CI > 10^{-4}$) at $0.5\times-4\times$ base capacity. Early layers stay stable; later layers absorb extras.

We train SPD at $0.5\times-4\times$ the base number of subcomponents and count how many stay "alive" (mean $CI > 10^{-4}$). When capacity grows, later layers absorb extras while early layers stay stable — SPD finds a **natural number of mechanisms per layer**. The parameter-faithfulness constraint prevents splitting by construction: components must sum to the original weights, so there is no room for redundant copies.

RESULT 4

Interpretability — Intruder Detection



Layer 3 MLP. SPD $CI>0.5$: 73.9% (c_fc), 66.1% (down_proj) vs. transcoder 61.9%. Random baseline = 20%.

Intruder detection measures interpretability: an LLM sees the top-activating examples for a feature plus one "intruder" from a different feature, and must identify the odd one out. Higher accuracy = more coherent, interpretable features. SPD components at $CI>0.5$ score **+12% absolute** over transcoder features.

TAKEAWAYS

Summary & Future Work

- Pareto-dominant:** SPD beats MSE-trained TCs and CLTs under all three LO normalizations.
- Protocol-robust:** Stable across eval modes; activation methods degrade 5–20× under mismatch.
- No feature splitting:** Parameter-faithfulness prevents splitting by construction.
- More interpretable:** +12% absolute on intruder detection vs. transcoders at comparable sparsity.

SPD accounts for network function in terms of the network's **actual objects doing the computation**.